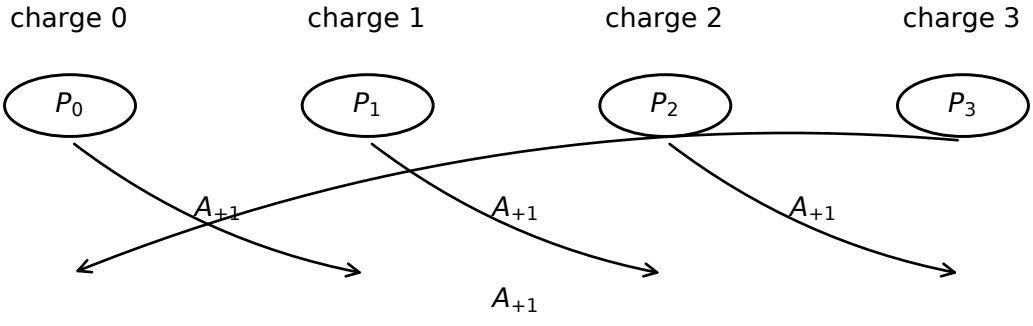

$SA_qS^* = \omega^q A_q$ forces $A_qP_j \subseteq P_{j+q}\mathcal{H}$